Qn	Solution
1(a)	gravitational potential energy at max. height $= mgh$ $27 = (0.43)(9.81)h$ $h = 6.4\ \text{m}$
1(b)	Applying conservation of energy, since horizontal velocity is constant, loss in KE due to vertical velocity = gain in GPE $\frac{1}{2}(0.43)v_y^2 = 27$ $v_y = 11\ \text{m s}^{-1}$
1(c)(i)	$v_y = u\sin 32° = 11\ \text{m s}^{-1}$ $v_x = u\cos 32° = \left(\frac{11}{\sin 32°}\right)\cos 32° = 17.6\ \text{m s}^{-1}$ Applying conservation of momentum in the horizontal direction, $(0.43)v_x = (0.43 + 0.67)v$ $v = 6.9\ \text{m s}^{-1}$
1(c)(ii)	The time for the ball to rise is equal to the time for it to fall. Since the horizontal velocity of the ball while it rises is $v_x = 17.6\ \text{m s}^{-1}$, it is greater than while it falls at horizontal velocity of $v = 6.88\ \text{m s}^{-1}$. Thus the horizontal distance travelled while it is rising is greater than when it is falling.
2(a)	Intermolecular forces between molecules is negligible.